

# **Experiment 01: Building Computer Network using Switch.**

## **Introduction**

Switches are essential networking devices used to connect multiple computers and devices within a Local Area Network (LAN). They operate at the data link layer to forward data frames based on MAC addresses, ensuring efficient communication within the network. Cisco Packet Tracer provides a virtual environment to design and simulate network connections without requiring physical hardware. In this lab, a network is created using switches to connect end devices and test their communication, helping students understand basic LAN configuration and data transmission concepts.

## **Objectives**

The main objectives of this lab are:

- To understand the role and functionality of a switch in a LAN.
- To design a simple network using a switch in Cisco Packet Tracer.
- To configure network devices with appropriate IP addresses.
- To verify connectivity between devices using commands such as ping.
- To gain hands-on experience in basic LAN setup and troubleshooting.

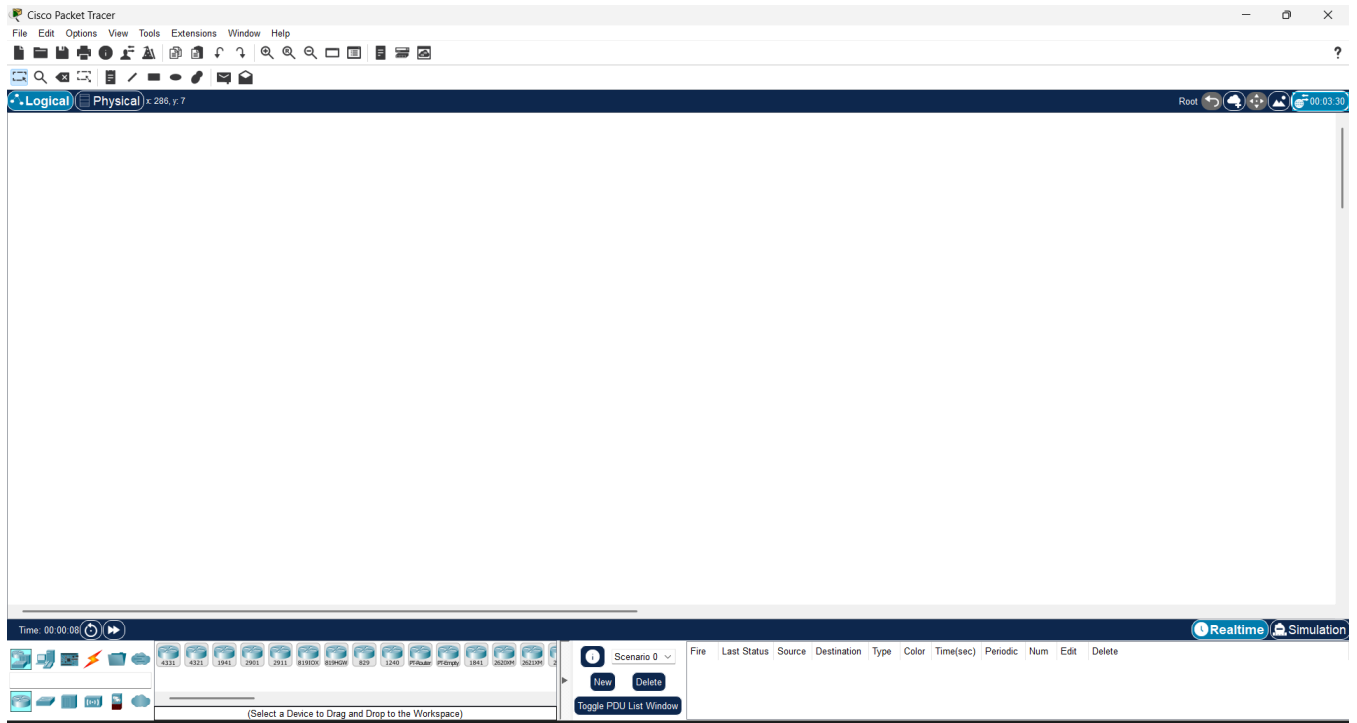
## **Equipment and Software:**

- **Software:** Cisco Packet Tracer (latest version)
- **Devices Used:**
  - 4 PCs (PC0, PC1, PC2, PC3)
  - 2 Switches (Switch0, Switch1)
  - 2 Routers (Router0, Router1)
  - Copper Straight-Through and Cross-Over Cables

## **Procedures of the Experiment:**

- **Opening Cisco Packet Tracer:**

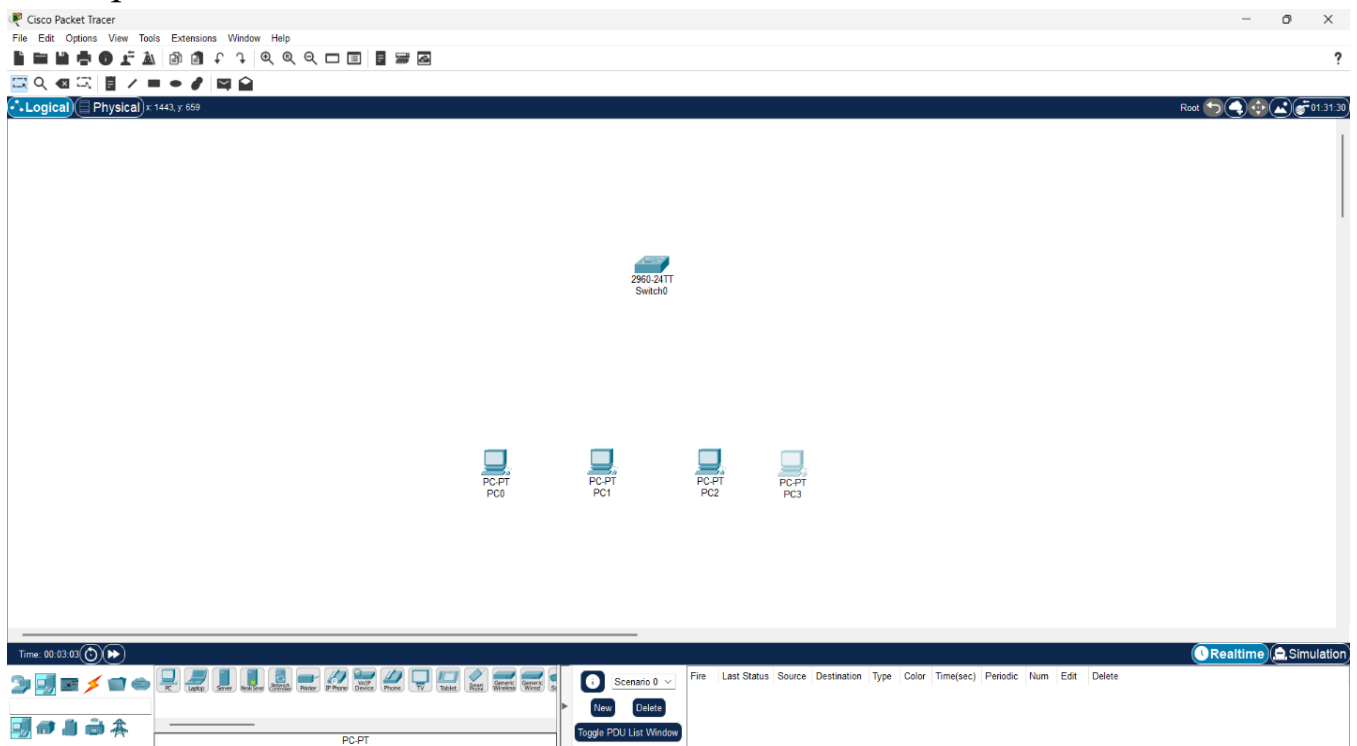
Launch the Cisco Packet Tracer Software and then open a new workspace on the software.



**Figure:1.1**

- **Adding Devices**

After opening, Drag and drop one switch (Switch0) and four PCs onto the workspace

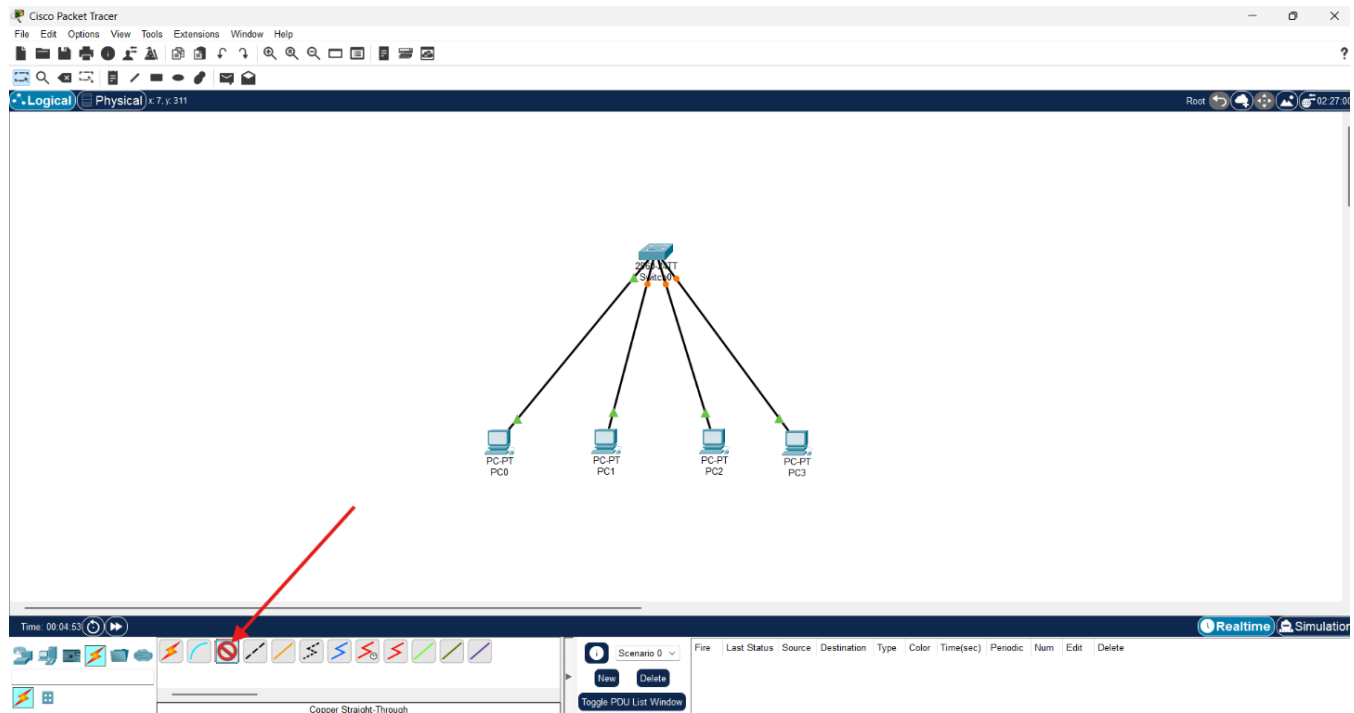


**Figure:1.2**

- **Connect Devices**

Use Copper Straight-Through Cables to connect each PC to the switch.

- PC0 → Switch0 (FastEthernet0/1)
- PC1 → Switch0 (FastEthernet0/2)
- PC2 → Switch0 (FastEthernet0/3)
- PC3 → Switch0 (FastEthernet0/4)

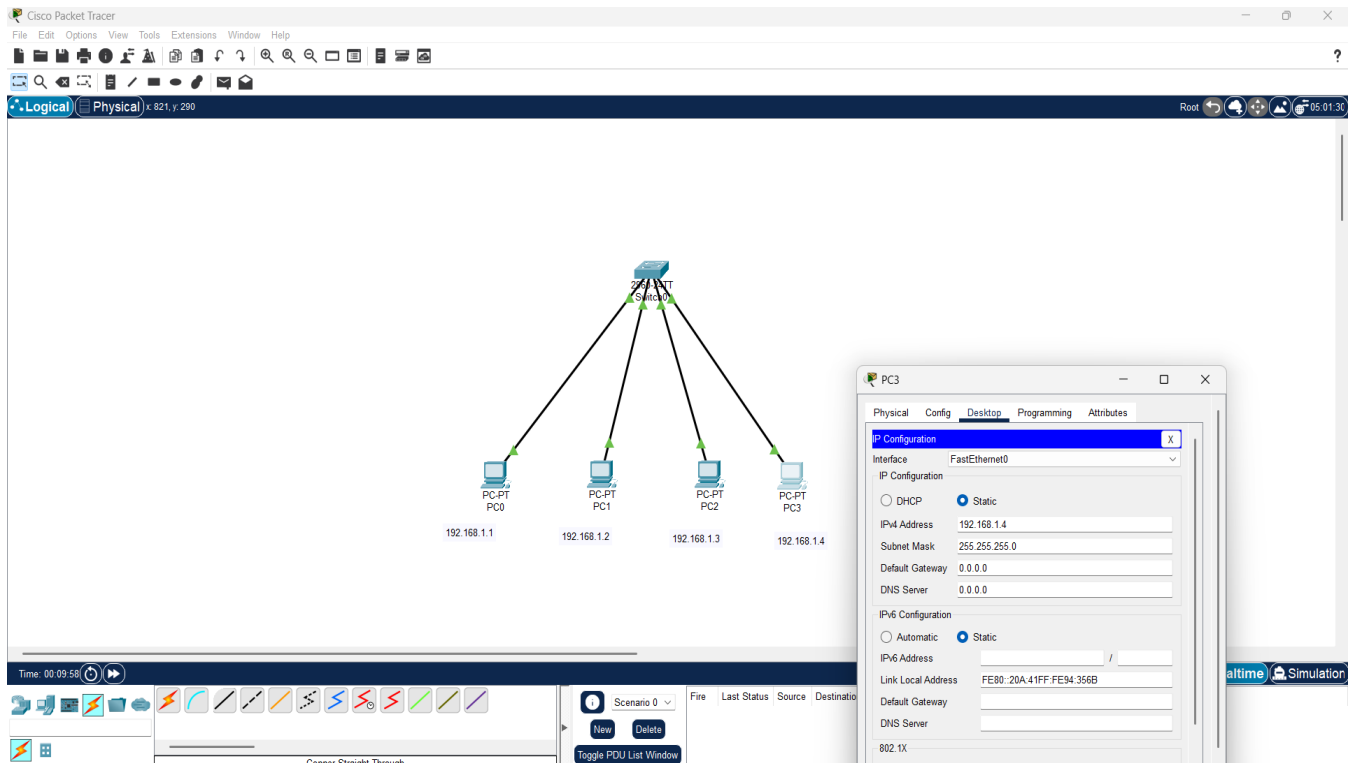


**Figure:1.3**

- **Assign IP Addresses**

Go to each PC → Desktop → IP Configuration, and assign the following:

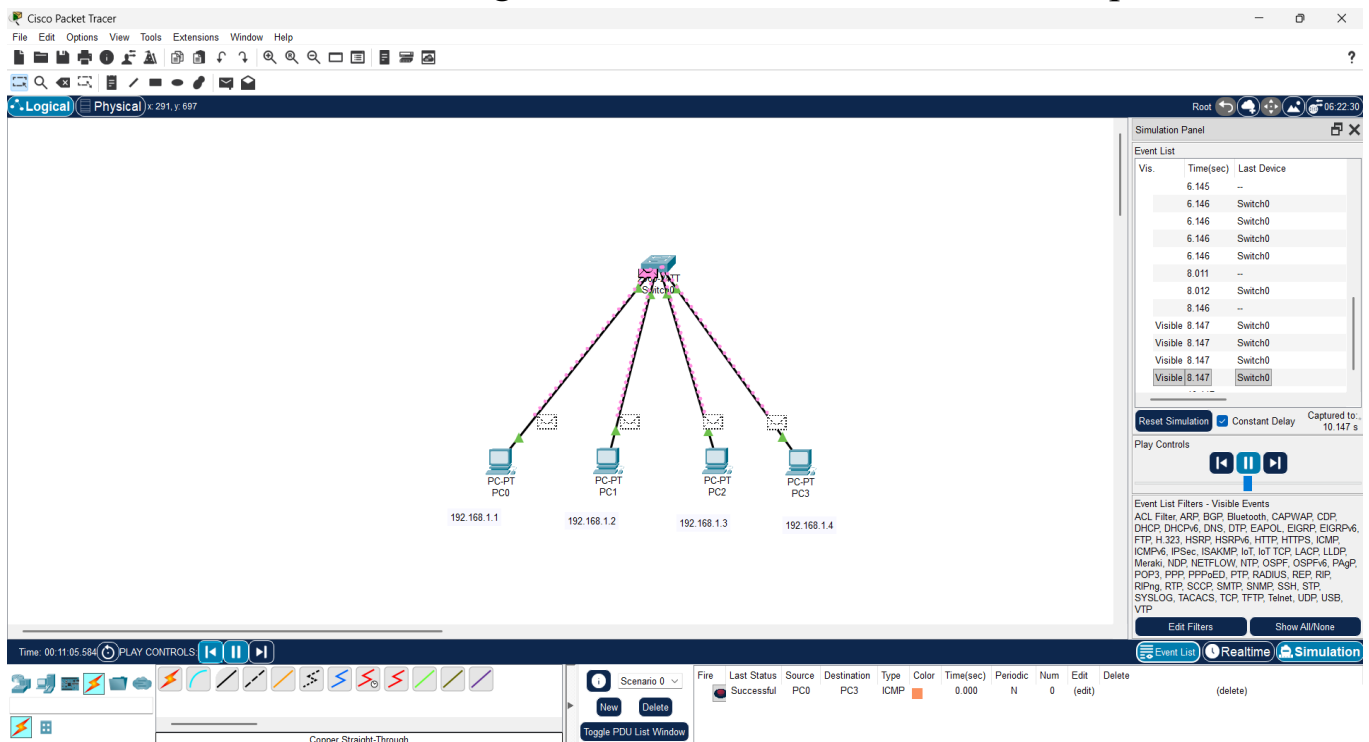
- PC0: 192.168.1.1 / 255.255.255.0
- PC1: 192.168.1.2 / 255.255.255.0
- PC2: 192.168.1.3 / 255.255.255.0
- PC3: 192.168.1.4 / 255.255.255.0



**Figure:1.4**

- **Test Connectivity**

Test The Connectivity By sliding the ICMP packets in each PC and then running the simulation. Here is the figure of the Successful Network setup.



**Figure:1.5**

- **Result**

All PCs were able to communicate successfully with each other through the switch. The switch efficiently created a local area network (LAN).

- **Conclusion**

This part demonstrated how to design and configure a simple LAN using a switch. The experiment verified the role of a switch in connecting multiple devices within the same network.